

Target validation paths — osteosarcoma-mets-dll3-h7r2

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

DLL3 IHC (clone SP347) is the test the case hinges on. A positive result at the trial threshold opens the only within-scope DLL3-directed therapy on the page (tarlatamab via NCT06788938). A negative result forecloses the entire DLL3-directed pathway — every BiTE, ADC, radioligand, and cell-therapy program in the dossier gates on protein-level confirmation, not transcript expression.

DLL3 RNA expression

Before any therapy decision: DLL3 IHC SP347 on archival FFPE, $\geq 1\%$ (preferably $\geq 25\%$) per NCT06788938's enrollment threshold. Turnaround is one to three weeks; cost is trivial relative to a treatment cycle. The first practical step is confirming SP347 assay availability at the treating institution — not every reference lab carries the Roche Tissue Diagnostics clone used in the tarlatamab development program.

Two refinements sit one tier down. Spatial heterogeneity is a known confounder in solid-tumor DLL3 (Zhang 2023): if multiple tumor blocks are available, IHC on a metastatic site as well as the primary refines confidence in whether the gating result generalizes. Neuroendocrine context (ASCL1 / NEUROD1 / chromogranin / synaptophysin / INSM1 panel) is research-grade for an osteosarcoma — DLL3 is normally a Notch-pathway target on neuroendocrine lineage, and an unexpectedly positive DLL3 IHC in a non-NEC tumor is worth contextualizing before the trial enrollment paperwork.

Germline TP53 sequencing (Li-Fraumeni panel) is a separate kind of finding — it does not affect tarlatamab eligibility, but late-teens / twenties osteosarcoma carries a meaningful prior probability for germline TP53. A positive result reshapes radiation-sensitivity considerations, screening for synchronous tumors, and family screening. Discuss with a genetic counselor before testing.

Where to order these assays

Assay	Provider	Contact
DLL3 IHC (clone SP347)	Foundation Medicine (FoundationOne CDx + IHC reflex)	test info · 1-888-988-3639
DLL3 IHC (clone SP347)	Mayo Clinic Laboratories	test info · 1-800-533-1710
DLL3 IHC (clone SP347)	NeoGenomics Laboratories	test info · 1-866-776-5907
DLL3 IHC (clone SP347)	Caris Life Sciences	test info · 1-888-979-8669
DLL3 IHC (clone SP347)	LabCorp / Esoterix Oncology	test info · 1-800-345-4363
Neuroendocrine IHC panel (ASCL1 / NEUROD1 / chromogranin / synaptophysin / INSM1)	ARUP Laboratories	test info · 1-800-242-2787
Neuroendocrine IHC panel	Mayo Clinic Laboratories	test info · 1-800-533-1710
Neuroendocrine IHC panel	LabCorp / Esoterix Oncology	test info · 1-800-345-4363
Neuroendocrine IHC panel	Quest Diagnostics	test info · 1-866-697-8378
Germline TP53 (Li-Fraumeni panel)	Invitae (Common Hereditary Cancers Panel)	test info · 1-800-436-3037
Germline TP53 panel	GeneDx	test info · 1-888-729-1206

Germline TP53 panel	Ambry Genetics (CancerNext)	test info · 1-866-262-7943
Germline TP53 panel	Myriad Genetics (MyRisk Hereditary Cancer)	test info · 1-800-469-7423

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.